Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 4.1 · Two secants

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 - 1$. Let A be the point of its curve with x = 1.

1. Calculate the slope of the secant joining A to the point on the curve with x-coordinate 2.

2. Repeat the calculation with the point of the curve with x-coordinate 1 ,1.

3. After studying the derivative at a point, determine $f\prime(1)$ and compare this value with the two rates calculated. Explain what changes when the second point approaches A.

Exercise 4.2 · One tangent

Consider the function $f(x) = 2x^2 + 1$, defined on $\mathbb{R}$. The aim is to determine the equation of its tangent at the point with x-coordinate -1, starting from the rate of change.

1. For $h \neq 0$, write then simplify $\dfrac{f(-1+h) - f(-1)}{h}$.

2. Deduce $f\prime(-1)$. Also calculate $f(-1)$.

3. Write the equation y = mx + p of the tangent and check that it passes through the point of contact.

Exercise 4.3 · Marginal cost

A company uses the model $C(q) = q^2 + 20q + 100$, where $q \geq 0$ denotes the quantity produced and $C(q)$ its cost in monetary units. The model extends the quantity to a real variable in order to study small variations in production.

1. Calculate $C(10)$ and $C(11)$, then the rate of change of C between 10 and 11. Interpret this rate in the context.
2. From the rate between 10 and 10 + h, determine $C\prime(10)$.
3. Compare the two results. Explain why the cost of one extra unit is not exactly equal to the derivative in this model.

Exercise 4.4 · Read a tangent

This exercise has two independent situations. A tangent to a curve is a line: another point on this line does not necessarily lie on the curve.

1. On the figure in the section on the tangent, read the coordinates of two points on the green line. Deduce its slope and check the value stated in the caption.
2. In another situation, the tangent to the graph of a function f at the point $A(2; 3)$ passes through $B(4; 2)$. Determine $f(2)$ and $f\prime(2)$. Can we deduce $f(4)$?
3. Use this tangent to give an approximate value of $f(2{,}1)$. Show the sign " approximately equal" in the conclusion.

Exercise 4.5 · Checking an approximation

We want to approximate $1{,}98^2$ without first carrying out the multiplication. We use the tangent to the curve of the square function at the point with x-coordinate 1.

1. Write the equation of this tangent and deduce the desired approximation.
2. Then calculate the exact value of $1{,}98^2$. The error is defined as "exact value minus approximate value": calculate this error and state its sign.
3. Explain the result from the identity $(1 + h)^2 = 1 + 2h + h^2$.

Exercise 4.6 · Tangent and intersection

The function f is defined on $\mathbb{R}$ by $f(x) = x^3$. We assume that $f\prime(1) = 3$. A student claims that a tangent and a curve always have exactly one point in common.

1. Determine the equation of the tangent to the graph of f at the point with x-coordinate 1.
2. Check by expanding that $x^3 - 3x + 2 = (x - 1)^2(x + 2)$.
3. Determine all points of intersection of the curve and this tangent. Is the pupil's claim correct?

Exercise 4.7 · Numerical secants

To study the square function near 1, we define, for $h \neq 0$,

$$T(h) = \frac{(1+h)^2 - 1}{h}.$$

Use the six values 0 ,1, 0,01, 0,001 and their opposites.

1. Write a Python function that calculates $T(h)$, then produce a table of six results.
2. Conjecture the value of the derivative of the square function at 1.
3. Simplify algebraically $T(h)$ to justify the conjecture. Explain why testing numerical values is not enough, on its own, to prove the result.

Exercise 4.8 · Delayed review

The reciprocal function is defined by $f(x) = \frac{1}{x}$ for $x \neq 0$. It is studied near 1.

1. State the forbidden values of h in the rate $\frac{f(1+h) - f(1)}{h}$, then simplify it.
2. Deduce $f'(1)$ and the equation of the tangent at 1.
3. Use this tangent to approximate $\frac{1}{1{,}02}$. Keep the distinction between exact value and approximate value.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 17 p. 127, all questions. Read two points on each tangent; write the difference quotient rather than estimating its steepness.

CQFD, exercise 19 p. 127, questions 1 and 2. Distinguish the y-coordinate of the point of contact from the slope of the tangent line.